CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 12: Algebraic Identities

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Expand $(2a + 7b)^2$.
2. Expand $\left(3x - \frac{1}{3x}\right)^2$.
3. Find the product $(7x - 4y)(7x + 4y)$.
4. Find the product $(x - 5)(x - 8)$.
5. Evaluate 206^2 and 198×202.
6. If $a + b = 10$ and $ab = 21$, find $a^2 + b^2$.
7. If $x + \frac{1}{x} = 5$, find $x^2 + \frac{1}{x^2}$.
8. Expand $(a - 2b - 3c)^2$.
9. Expand $(3x - 2)^3$.
10. If $a - b = 2$ and $ab = 15$, find $a^3 - b^3$.

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Answer Key

1. $4a^2 + 28ab + 49b^2$
2. $9x^2 - 2 + \frac{1}{9x^2}$
3. $49x^2 - 16y^2$
4. $x^2 - 13x + 40$
5. 42436 and 39996
6. 58
7. 23
8. $a^2 + 4b^2 + 9c^2 - 4ab + 12bc - 6ca$
9. $27x^3 - 54x^2 + 36x - 8$
10. $8 + 90 = 98$

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